# Continuous Probability Distributions

GENERAL ASSEMBLY

# Agenda

- Review of random variables and discrete distributions
- Continuous distribution rapid fire
- Continuous distributions in Python

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# Recap

# Earlier...

We studied **discretely distributed random variables**.



In your own words, what is a **random variable**?

# Earlier...

We studied **discretely distributed random variables**.



In your own words, what is a **distribution**?

# Rapid Fire: What are each of these?

- Uniform distribution
- Bernoulli distribution
- Binomial distribution
- Poisson distribution
- Brownie points: Any others?

# The Differences

# Density Plots

For discrete variables, we had a **probability mass function (pmf)** that said “at a given value, what is its probability occurring?”

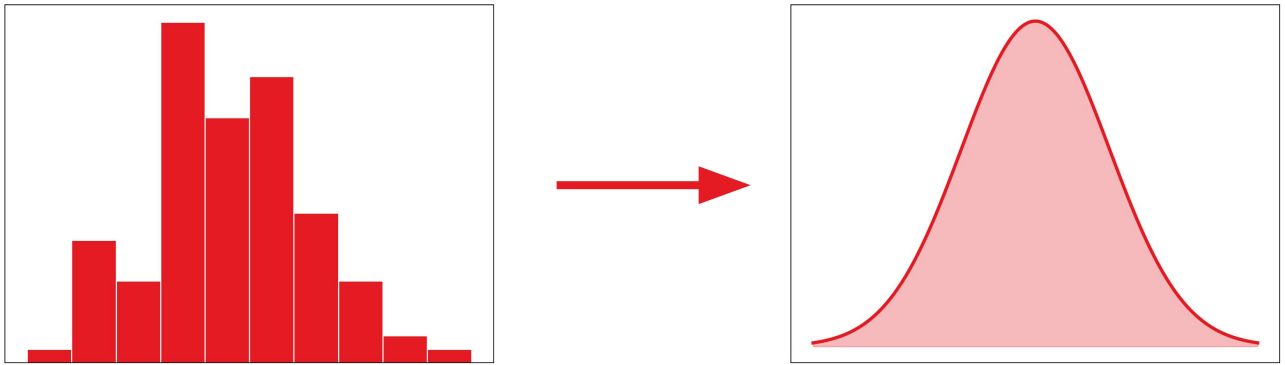
We don’t have such a thing for continuous variables.

***Why?***

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# Density Plots

Instead of a pmf, we have a **probability density function (pdf)**. This is a function that describes the comparative probability of an event occuring. Since we can't find probabilities at a given point, **we find probabilities along intervals**.



# Calculus Aside

If you've taken a calculus course before, this might sound like an **integral** to you. Surprise: it is!

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

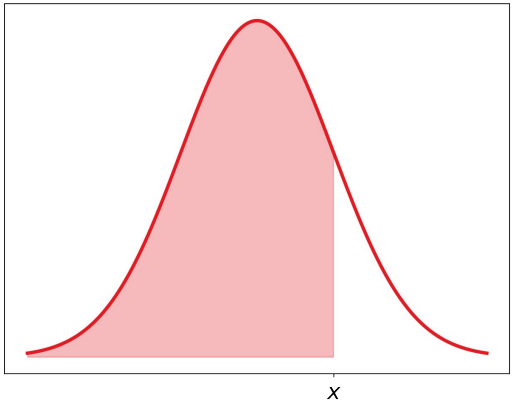
Of course, this is ***not*** a calculus class, so you won't be doing this by hand. Lucky you!

# CDFs

Because of the calculus trickery, a lot of continuous probability is told via **cumulative density functions (CDFs)**. Simply put, a CDF of a distribution is the probability of getting at or below a certain value. Mathematically:

$$F(x) = P(X \leq x)$$

Pretty much all the “named” distributions have their pdfs and CDFs made for us in the SciPy library.

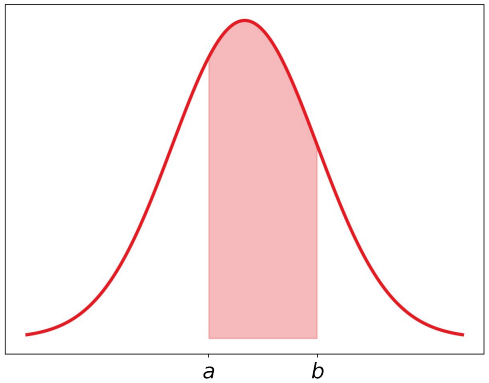



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# Probabilities via CDFs

This is typically how we compute probabilities of continuous functions. There's a lot on this slide, so let's just soak it in:

$$P(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$$



# The Common Continuous Distributions

# The Common Continuous Distributions

On the next several slides, we'll be talking about some of the more major, well-known continuous distributions. This isn't all of them, of course. These are just the ones you'll see the most, and you'll see them all again later:

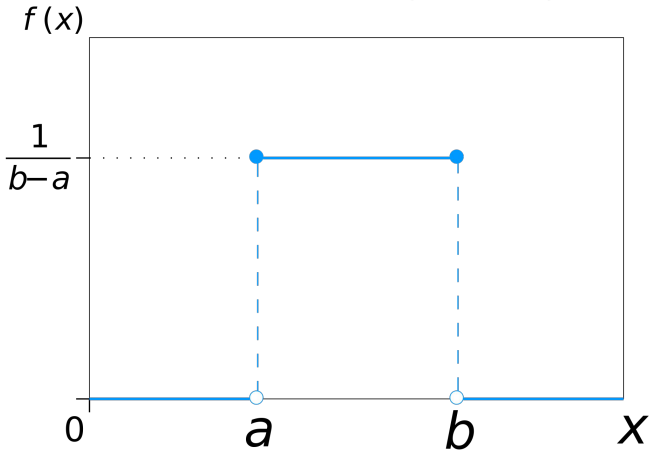
- Continuous Uniform
- Normal (and the Standard Normal)
- t
- Exponential
- Gamma
- Beta

# The Continuous Uniform

If all numbers in a *continuous* interval [*a*, *b*] are equally likely, you could use a **continuous uniform distribution** to model this scenario.

| | |
|---|---|
| **pdf** | $f(x) = \frac{1}{b-a}$ |
| **support** | $a \leq x \leq b$ |
| **mean** | $\frac{a+b}{2}$ |
| **variance** | $\frac{(b-a)^2}{12}$ |

$$X \sim U(a, b)$$




Stolen unapologetically from Wikipedia.

# The Normal Distribution

Maybe the most well-known is the **normal distribution** (sometimes known as the **Gaussian distribution**). It is bell-curved, symmetric, and entirely described by its mean and variance $\mu$ and $\sigma^2$.

We typically **assume** something is normally distributed when we have a continuous number that varies from some mean, equally likely above and below.

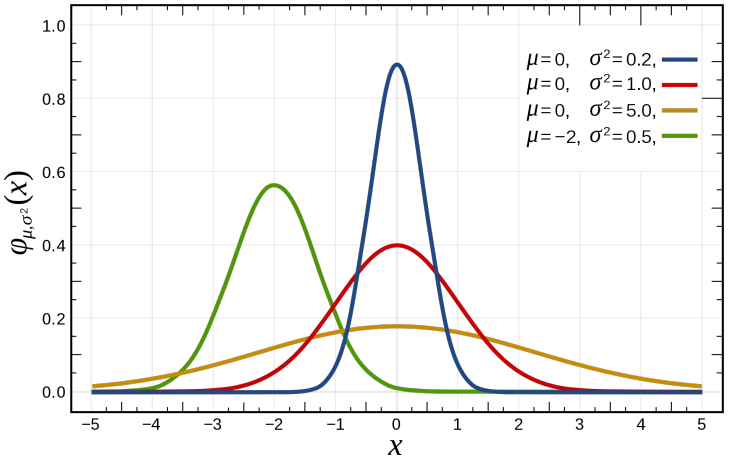
While it's actually surprisingly **rare** for something to *really* be normally distributed, it makes a very good approximation in a lot of real scenarios. It's also a very fundamental distribution to a lot of statistical theory.

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# The Normal Distribution

| | |
|---|---|
| **pdf** | $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$ |
| **support** | $-\infty < x < \infty$ |
| **mean** | $\mu$ |
| **variance** | $\sigma^2$ |

$$X \sim N(\mu, \sigma^2)$$




Stolen unapologetically from Wikipedia.

# The Normal Distribution

This is so important, we should spend more time playing with this:

https://digitalfirst.bfwpub.com/stats_applet/stats_applet_7_norm.html

# The Standard Normal Distribution

There's a special case of the normal distribution called the **standard normal distribution**, which occurs when $\mu = 0$, $\sigma = 1$. It can be attained by standardizing the normal distribution, just like we did when standardizing data the other day!

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

$$\implies Z \sim N(0, 1)$$

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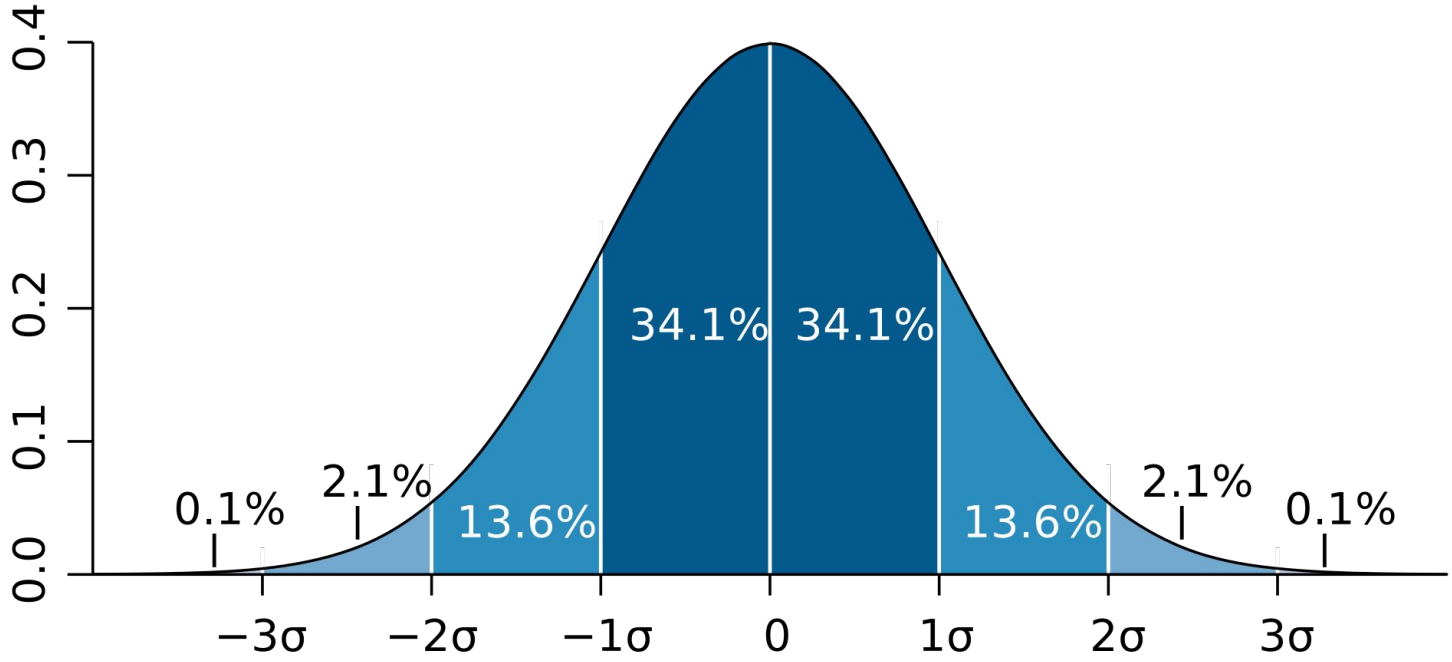
# The Empirical Rule

The **Empirical Rule** (or the **68-95-99.7 Rule**) is a commonly used guideline for thinking about probabilities. It states the following about the **(standard) normal distribution:**

- The probability of an event occurring within 1 standard deviation from the mean (ie, $-1 \leq z \leq 1$) is **68%**.
- The probability of an event occurring within 2 standard deviation from the mean (ie, $-2 \leq z \leq 2$) is **95%**.
- The probability of an event occurring within 3 standard deviation from the mean (ie, $-3 \leq z \leq 3$) is **99.7%**.

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# The Empirical Rule, Visualized



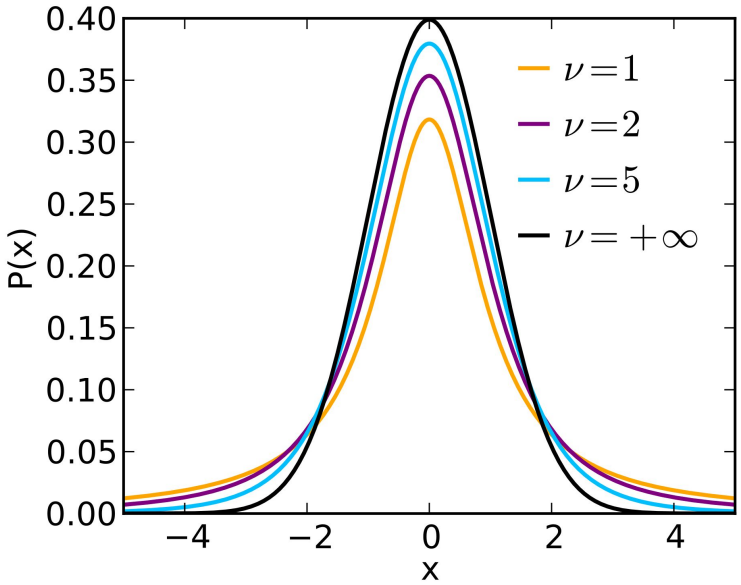

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# Student's *t*-distribution

This is a weird one. Graphically, it looks a *lot* like the standard normal.

However, it has an additional parameter: its **degrees of freedom**, usually denoted *df* but sometimes as the Greek letter $\nu$.




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# Student's *t*-distribution

In fact, as the number of degrees of freedom increases, it approaches the standard normal asymptotically. That is,

$$\lim_{\nu \to \infty} T_\nu = Z$$

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# Student's *t*-distribution

What's this for? Inference!

Soon, we're going to encounter scenarios where we want to **find** the unknown true population $\mu$, and we're going to estimate it by taking samples. However, in order to use the normal distribution, we'd need to know the true population SD, $\sigma$. That's kinda nonsense, right? When would we ever know $\sigma$ but not $\mu$?

The *t*-distribution allows us to use the sample SD instead of the population SD to do this sort of work. Keep it in mind!

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# Aside: William Sealy Gosset (aka Student)






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# The Exponential Distribution

The **exponential distribution** is a heavily right-skew distribution. It is very commonly used to model **waiting-time** data.

It has only one parameter, $\lambda$... or is it $\square$?

# The Exponential Distribution

Suppose Tim gets a spam email every 10 hours. You could model the time between spam emails via the exponential distribution with *rate* 1/10. Many resources instead prefer to just use this 10 as the parameter, since it also happens to be the mean. That is...

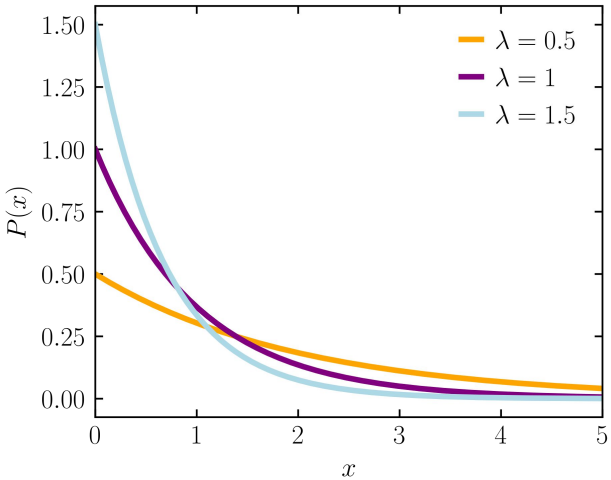
$$X \sim \text{Exp}\left(\frac{1}{10}\right) \quad \text{and} \quad X \sim \text{Exp}(10)$$

Are both commonly written to mean the same thing. Be mindful of the context! I prefer the latter (mean notation) rather than the former (rate notation).

# The Exponential Distribution (mean notation)

| | |
|---|---|
| **pdf** | $f(x) = \frac{1}{\beta} e^{-x/\beta}$ |
| **support** | $x > 0$ |
| **mean** | $\beta$ |
| **variance** | $\beta^2$ |

$$X \sim \text{Exp}(\beta)$$

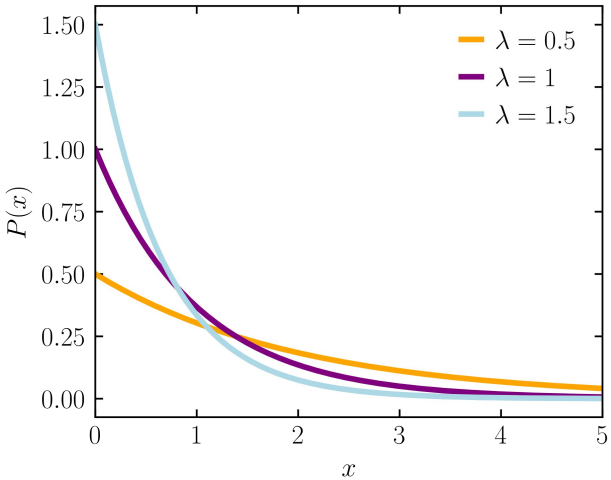



Stolen unapologetically from Wikipedia.

# The Exponential Distribution (rate notation)

| | |
|---|---|
| **pdf** | $f(x) = \lambda e^{-\lambda x}$ |
| **support** | $x > 0$ |
| **mean** | $\frac{1}{\lambda}$ |
| **variance** | $\frac{1}{\lambda^2}$ |

$$X \sim \text{Exp}(\lambda)$$




Stolen unapologetically from Wikipedia.

# Exponential and Poisson

The exponential distribution and the Poisson distribution have an interesting relationship. Remember:

- The Poisson is often used to model how many times an event occurs in a fixed timeframe
- The exponential is often used to model how long it takes for an event to occur
- So, the time between Poisson events is exponentially distributed!
- Modeling both of these things in conjunction is called a **Poisson process** and is a topic of study in **stochastic statistics** and is way beyond our scope. But it's cool, though!

# The Gamma Distribution

I won't spend a lot of time on this. It's very similar to the exponential distribution except with an additional **shape** parameter, $\alpha$. I'll include only two interesting gamma facts that are worth knowing at this time:

- Just like exponential, it has support x > 0, is skew right, and is used to model waiting times.
- ... Actually, it's used to model $\alpha$ consecutive waiting times! That is... if we have $\alpha$ exponentials:

$$X_1 + \cdots + X_\alpha \sim \text{Gamma}(\alpha, \beta)$$

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# The Beta Distribution

This one is also very strange, and we won't see it too often. So, I'll keep the interesting facts limited:

- Has two parameters, $\alpha$ and $\square$, and in practice they're usually the same number. They control how "bowl" or "hill" shaped the pdf is.
- Since the support is between 0 and 1, the Beta distribution is often used to model probabilities in **Bayesian statistics**, which we'll get to much later.
- The standard uniform is a special case: Beta(1, 1) = U(0, 1)
- We can take a look at some cool Beta pics here.

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# Finale

# RV Relationships

You may have noticed there's a lot of relationships between various random variables. You have touched only the tip of the iceberg, my friend.

Check this out.

# Bonus: Multivariate Distributions

Statistics is a wide-open field, and we're really only learning what's prerequisite for the major data science topics in our course. There are also **multivariate distributions** that are represented by vectors instead of scalars. I'll briefly mention two interesting ones:

- The **multinomial distribution**: What if a trial had more than two options?
- The **multivariate normal distribution**: What if you're generating a vector of values from a normal distribution... but they're not independent?